STATE ESTIMATION OF A 2D TURNING OBJECT WITH EXTENDED KALMAN FILTER

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Abstract

This report studies the state estimation of a coordinated turn model with mean-reverting speed and turn rate. The mean-reversion prevents drifts in speed and turn rate. The states are estimated using extended Kalman filter with adaptive noise covariance. Three sensors provide nonlinear measurements using range-only information. To improve on the stability, The simulation results show high accuracy even in the indirectly observed states.

1 Introduction

In a coordinated turn model the target's state is comprised of position (x_1, x_2), speed s, heading ψ, and turn rate ω. When the state is represented in polar coordinates, the sin and cosine functions make the dynamics non-linear (Zhou and Guo, 2019). Tracking the trajectory and estimating the states then requires the use of the extended Kalman filter.

This project uses a mean-reverting spiral, where the object's angular velocity and speed are stochastic processes. The objective is to implement the extended Kalman filter to estimate states accurately.

2 Model

A polar coordinated turn model has a dynamic model

$$x_k = f(x_{k-1}) + q_k \tag{1}$$

where the state vector is $x = [x_{1,k}, x_{2,k}, s_k, \phi_k, \omega_k]$. For this problem, we consider the polar velocity kinematics

$$dx = \begin{pmatrix} s\cos(\psi) \\ s\sin(\psi) \\ \theta_s(\mu_s - s) \\ \omega \\ \theta_\omega(\mu_\omega - \omega) \end{pmatrix} dt + \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \sigma_s & 0 \\ 0 & 0 \\ 0 & \sigma_\omega \end{pmatrix} dW \tag{2}$$

The coefficients θ_s and θ_ω determine the pull at which the speed and turn rate revert to their long-term means θ_s and μ_ω. Together with the Wiener increments, the state variables are constrained and the spiral pattern stays stable as the unbounded drift of a simple random walk is

prevented.

The stochastic differential equation is then discretized using the Euler-Maruyama method

$$\begin{cases} x_{1,k} = x_{1,k-1} + s_{k-1} \cdot cos(\psi_{k-1})\Delta t \\ x_{2,k} = x_{2,k-1} + s_{k-1} \cdot sin(\psi_{k-1})\Delta t \\ s_k = s_{k-1} + \theta(\mu_s - s_{k-1})\Delta t + \sigma_s \sqrt{(}\Delta t)\epsilon_{k-1}^{(s)} \\ \psi_k = \psi_{k-1} + \omega_{k-1}\Delta t \\ \omega_k = \omega_{k-1} + \theta_\omega(\mu_\omega - \omega_{k-1})\Delta t + \sigma_\omega \sqrt{(}\Delta t)\epsilon_{k-1}^{(\omega)} \end{cases} \tag{3}$$

The dynamic model function is then expressed as

$$f(x_{k-1}) = \begin{pmatrix} x_{1,k-1} + s_{k-1} \cdot cos(\psi_{k-1})\Delta t \\ x_{2,k-1} + s_{k-1} \cdot sin(\psi_{k-1})\Delta t \\ s_{k-1} + \theta_s(\mu_s - s_{k-1})\Delta t \\ \psi_{k-1} + \omega_{k-1}\Delta t \\ \omega_{k-1} + \theta_\omega(\mu_\omega - \omega_{k-1})\Delta t \end{pmatrix} \tag{4}$$

The transition matrix is then defined as $F = \frac{\partial f}{\partial x}|_{m_{k-1|k-1}}$. The measurement matrix is obtained with the equation $H = \frac{\partial h}{\partial x}|_{m_{k|k-1}}$ (Simo Särkkä, 2013).

Measurements are made by three sensors at positions $[(s_{1,x}, s_{1,y})(s_{2,x}, s_{2,y}), (s_{3,x}, s_{3,y})]$. The measurement model is

$$y_k = \begin{pmatrix} \sqrt{x_{1,k} - s_{1,x} + x_{2,k} - s_{1,y}} \\ \sqrt{x_{1,k} - s_{2,x} + x_{2,k} - s_{2,y}} \\ \sqrt{x_{1,k} - s_{3,x} + x_{2,k} - s_{3,y}} \end{pmatrix} + \begin{pmatrix} r_{1,k} \\ r_{2,k} \\ r_{3,k} \end{pmatrix} \tag{5}$$

The measurement noise $q_k \sim N(0, Q)$ and $r_k \sim N(0, R)$.

3 Methods

Each sensor measures the distance to the moving object. These sensors are fixed to their respective position. Three sensors are is the minimum required of number of sensors. Less sensors could introduce collinearity and as consequence the exact location becomes ambiguous. Even in the case of three sensors, the third sensor must not be collinear. (Sadaphal and Jain, 2009). The extended Kalman filter prediction and update equations are

predict:

$$\begin{aligned} m_{k|k-1} &= f(m_{k_1|k-1}) \\ P_{k|k-1} &= F_x(m_{k-1})P_{k-1|k-1}F_x^T(m_{k-1}) + Q_{k-1} \end{aligned}$$

update:

$$\begin{aligned} v_k &= y_k - h(m_{k|k-1}) \\ S_k &= H_k P_{k|k-1} + R_k \\ K_{k|k} &= P_{k|k-1}H_k^T S_k^{-1} \\ m_{k|k} &= m_{k|k-1} + K_k v_k \\ P_{k|k} &= (I - K_k H_k)P_{k|k-1} - (I - K_k H_k) + K_k R_{k-1} K_k^T \end{aligned}$$

(*Groves*, 2013). (6)

In order to make the filter more robust, the measurement noise covariance is adjusted according the difference between theoretical errors and actual errors.

First the residuals are calculated

$$\varepsilon_k = v_k - h(x_{k|k)} \tag{7}$$

Then measurement noise covariance is updated

$$\hat{R}_k = \beta \hat{R}_{k-1} + (1-\beta)(\varepsilon_k \varepsilon_k^T + H_k P_{k|k-1} H_k^T) \tag{8}$$

The process noise is also adapted

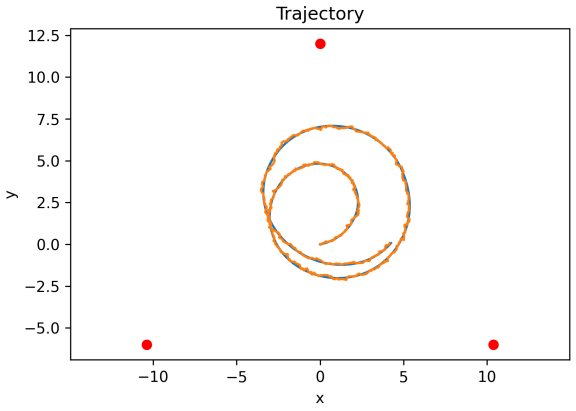
$$\hat{Q} = \beta \hat{Q}_{k-1} + (1-\beta)(K_k v_k v_k^T K_k^T) \tag{9}$$

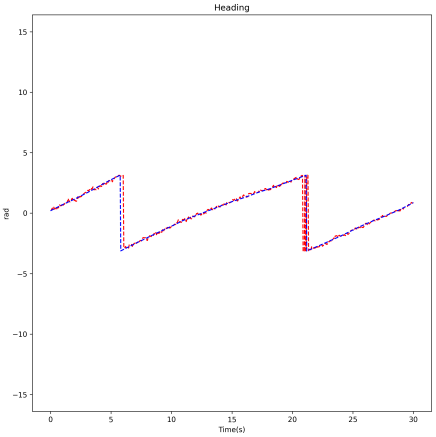
$\beta \in [0, 1]$ is the forgetting factor. (Akhlaghi et al., 2017). The noise values are accepted whenever they are within the bounds $[1e-6, 1e6]$ for R, and $[1e-8, 1e4]$ for Q.

4 Numerical Examples

RMSE (x_1, x_2)	RMSE s	RMSE ψ	RMSE ω
0.0886	0.202	4.983°	0.0597 rad/s

Table 1: RMSE of different states





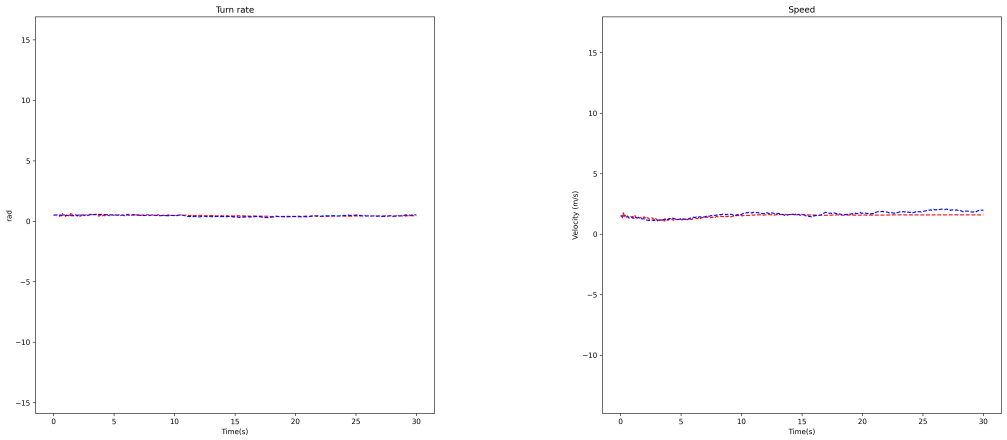


Figure 1: EKF results: Trajectory,heading , turn rate and speed.

5 Conclusions

The Extended Kalman filter with adaptive noise provide accurate results. Despite of the fact that velocity, heading and turn rate were only indirectly observable the error were smaller than expected. The tuning of covariance did improve robustness but this project could improved by considering the measurement angles. The curved motion of Coordinated Turn model enables indirect access to the velocity, heading and turn rate. The mean-reversion helped to bound the drift and reducing the variance. Both of these provided stability to the filter performance.

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